Project Scope, Objectives, and Literature Review Report

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Friday 4 April, 2025

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__Disclaimer__ – All these conference papers have been submitted as partial fulfilment for the project requirement for the BE(Hons) degree. Although they have been assessed, no errors or factual information have been corrected or checked.

ABSTRACT

Presented in this report, commissioned by the University of Auckland research department, is a literature review that investigates the requirements to develop a reconfigurable NPU architecture that dynamically adjusts computational precision and dataflow to enhance energy efficiency and maintain high throughput. To achieve such a goal, the research dives into the current technological efforts and innovations such as low-precision computation, sparsity aware processing adaptive precision switching etc, to minimise energy consumption. The potential of systolic arrays optimisation is also explored to reduce the memory access latency. For implementation, the proposed architecture will be prototyped on FPGAs, as they are extremely flexible for real-time configuration.

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1 PROJECT SCOPE AND OBJECTIVES

1.1 Introduction & Problem Statement

Machine learning (ML), a branch of Artificial Intelligence, uses algorithms that learn from data to perform specific tasks, continuously improving without being programmed for every task. They are used in applications such as image processing and speech recognition, making them extremely useful in today's ever-growing technological society. Although these machine learning algorithms are favourable, running them on the edge may cause issues such as high computational power and energy consumption. This is when hardware accelerators such as Neural Processing Units (NPU) were invented and specialised for machine learning, improving energy consumption and producing high throughput. However, existing NPUs are fixed in terms of operation and lack flexibility since they are only designed for specific tasks. As the complexity of ML models is rapidly increasing, a reconfigurable NPU is necessary to produce a more energy-efficient outcome while increasing system performance.

This research aims to investigate the computational complexity of machine learning algorithms, especially, deep learning algorithms for embedded applications and develop a reconfigurable hardware accelerator, implemented on FPGA to increase performance and minimise energy consumption for devices at the edge.

This literature review report dives into the current advancements and limitations in the designing and implementation of energy-efficient Neural Processing Units, with a focus on systolic array optimisation and FPGA implementation. By reviewing several recent research efforts, this review identifies existing gaps for further investigation.

1.2 Objectives

- Design an NPU architecture that can dynamically adjust it's configuration that balances both energy efficiency and computational performance
- Explore methods that increase the energy efficiency of systolic arrays, focusing on sparsity-aware techniques.
- utilise the reconfigurable nature of FPGAs to prototype NPUs and focus on achieving high computational throughput
- Benchmark the NPU against existing architectures to measure the efficiency and throughput in edge-based applications

1.3 Research Scope

The research project is covered over 24 weeks, where the aim is to achieve an operating iteration of a reconfigurable NPU implemented on an FPGA using hardware description language with dynamic partial configuration. The reconfigurable NPU will be designed with systolic array optimisation as its core component. Testing the energy efficiency will be restricted to CNNs, which the development of will not be included in the research. For the FPGA implementation, it will be on an intel cyclone V DE1-SoC due to the integrated ARM processor and high DSP count, perfect for configuring adaptive real-time management system and systolic array optimisation.

The responsibilities for the tasks laid out in the gantt chart are equal for all members of the project. However, specialised tasks will be delegated to ensure more coverage, such as each member can focus on a different method of increasing energy efficiency.

1.4 Gantt Chart

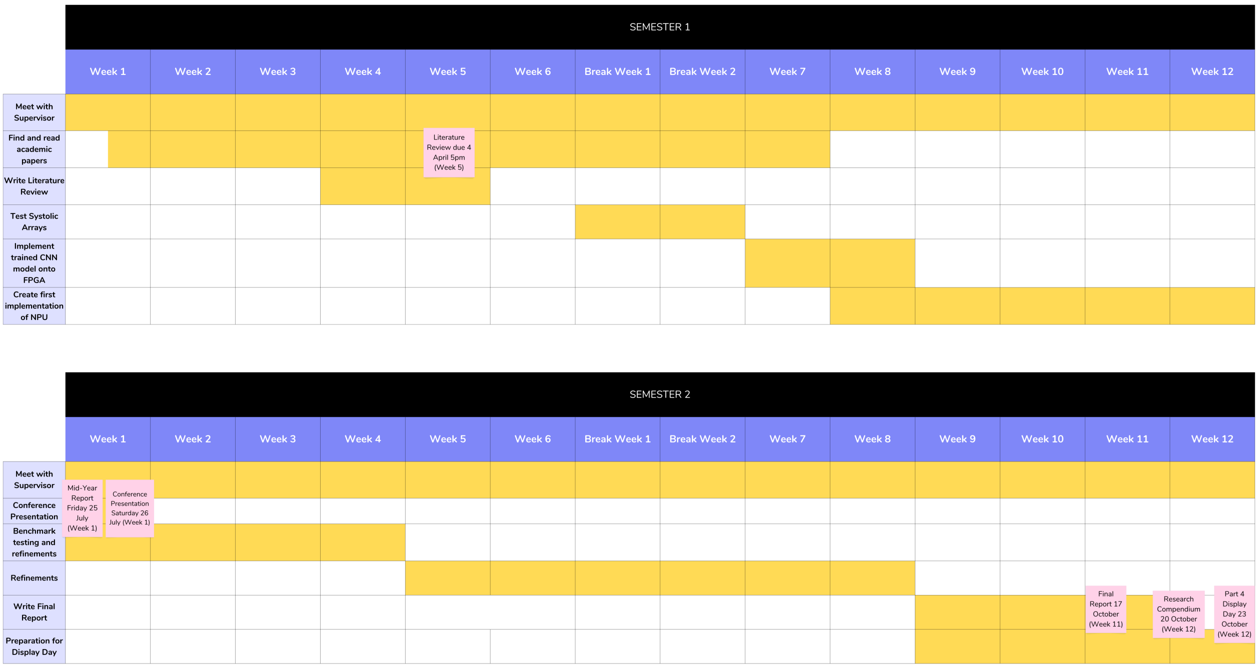

Figure 1: Gantt Chart for both semesters (24 weeks)

2 LITERATURE REVIEW

2.1 Background: Machine Learning

Machine Learning (ML) is a field in Artificial Intelligence (AI) that specialises in enabling computers to learn from data by utilising training algorithms so that they can make predictions based on data input. The most important sub-category within machine learning is deep learning, where machine learning utilises a concept called neural networks. Inspired by the brain, a neural network is a model of neuron-like nodes that process data by adjusting the weight and biases. A weight determines how significant the influence of the input is on the output, and a bias sets a default threshold value in order to define what the network will output when no proper decision has been made. Without bias, a neuron's activation would be tied directly to the inputs and cause ambiguity. The "processing" process is done within layers consisting of an input layer (input data), an output layer (output data) and a hidden layer, which involves all the neurons multiplied with a weight and added with a bias ("But what is a neural network? — Deep learning chapter 1 - YouTube", n.d.). Amongst all the types of different neural networks, the most commonly used one is a convolutional neural network (CNN).

2.1.1 Convolutional Neural Networks

A CNN consists of convolutional layers where a kernel is applied to perform convolution operations to extract important data and pooling layers, which help reduce the computational load. This particular neural network specialises in image analysis, such as object detection and facial recognition (Guha & Chakrabarti, 2024). CNNs face a particular challenge where they cannot take arbitrary input sizes, causing further issues when dealing with dynamic datasets. The research conducted by (K.S et al., 2022) attempts to work around that issue by combining a CNN with spatial pyramid pooling (SPP), enabling variable input sizes without jeopardising the performance. The model demonstrated by this research paper showed an energy efficiency of 85.9 bps/watt, outperforming conventional methods while maintaining a 95% accuracy.

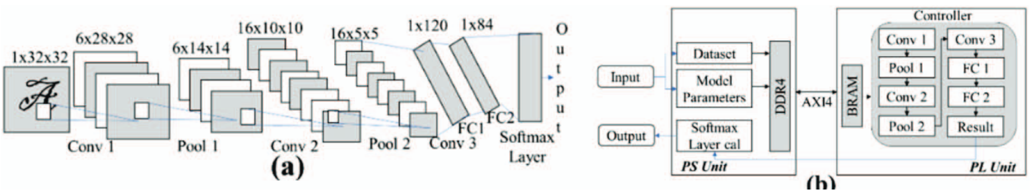

Figure 2: Visual Representation of LeNET5 CNN model (Guha & Chakrabarti, 2024)

2.2 Neural Processing Unit (NPU)

2.2.1 Context and Importance of an NPU

A neural processing unit is a specialised microprocessor and hardware accelerator for neural network computations. The integrated NPUs made by intel as shown in figure 3 have two Neural Compute Engines, each of which consists of two Digital Signal Processing Units (DSP) and an inference pipeline which consists of other units, the most important being the Multiply-Accumulate (MAC) array. The MAC array consists of processing elements (PEs) that perform the mathematical functions of multiplication and accumulation. These processing elements are organised as a grid, structured as a systolic array, where each processing element performs MAC operations simultaneously. This becomes extremely useful when it comes to high-speed matrix computations within convolutional layers in deep learning. It also contains a high bandwidth on-chip memory, allowing it to handle large amounts of data efficiently and run multiple neural network operations simultaneously. Neural networks require relatively quick access to the relevant metadata of the network, such as model weights, biases and inputs; this way, the NPU can perform millions of computations that access the data stored next to the processing cores. This makes the on-chip memory a necessary component as these millions of computations need to be processed whilst maintaining a high throughput and data retrieval. While traditional processors such as CPUs and GPUs are quite powerful, the current growth in machine learning demands for high speed processing and energy efficiency, thus making Neural Processing Units highly dependable for machine learning applications. An experiment from a recent study has shown results where NPU acceleration can provide a 2.3 times increase in speed 3 times the energy saving compared to CPU operation (Esmaeilzadeh et al., 2012).

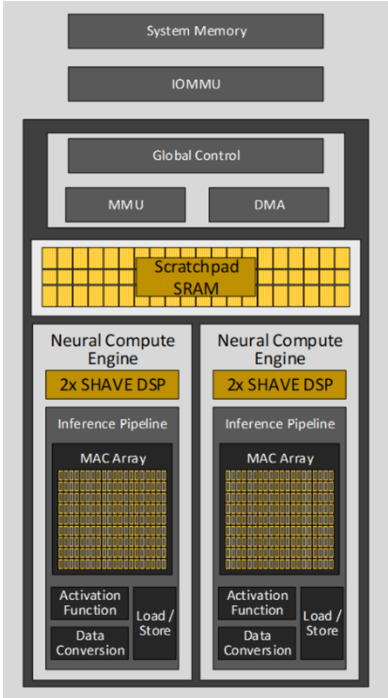

Figure 3: Intel based integrated NPU architecture
("Quick overview of Intel's Neural Processing Unit (NPU) — Intel® NPU Acceleration Library documentation", n.d.)

2.2.2 Current Challenges and Proposed Solutions

Despite having improved the Artificial Intelligence industry with the introduction of the NPU and it is remarkable parallel processing of neural networks and high throughput, they also face a few challenges. One of them being parallel computations. Although this is a positive quality as it allows the NPU to perform millions of calculations at a time, it also poses as a weakness as it can become a bottleneck when dealing with irregular parallelism, leading to processing elements unable to perform in lockstep (L. Liu & Brown, 2021). This can be caused by irregular workloads, breaking the rhythm as some PEs may complete the operation early and sit idle, which would waste computational resources. Another challenge is facing issues when memory read and write operations are required. Memory read and write operations have a high operating cost but sometimes become necessary due to complex computations, further worsening the bottleneck. (X. Liu et al., 2024) mentions Fast Fourier Transform (FFT) as an example where the NPU can have bottleneck issues as the non-processed data cannot be accessed continuously, which results in frequent memory access operations. The cost of memory operations is further increased as the FFT involves complex operations which needs to be split into real and imaginary parts, thus leading to higher computational costs. To combat these challenges, far more innovative and uncommon methods must be utilised that can allow the NPU to perform computation with regular parallelism whilst maintaining an efficient method of memory access.

- **Low-precision computation** is a method proposed in (Gorvadiya et al., 2025) shown to enhance energy efficiency by reducing memory usage and enabling more computation units within the same area. It does this by reducing the number of bits required to represent the weights of a neural network. Several forms of Quantisation have been explored, such as fixed-point Quantisation, where there is a fixed number of bits, e.g. 8-bits. This is compared to dynamic Quantisation, where the model in question is trained using a larger bit precision value and reduced further during the inference stage, improving the performance. Several data representations were also experimented with, such as INT8, FP16, and FP32. The 8-bit integer representations were observed to be the most promising, as memory requirements were significantly reduced. Whereas 16- and 32-bit floating point representations had higher accuracies, they were at the cost of computational resources, making INT8 the best choice for edge-based devices where energy efficiency is crucial. This is tested and proven with the results displayed in figure 4 where 8-bit representation for Quantisation reduced memory usage by up to 75% while still maintaining plausible accuracy. Despite the benefits demonstrated, in the context of edge-based reconfigurable NPUs, there may be a potential loss of accuracy, especially when dealing with sensitive layers within neural networks, since different layers within a network require different precision levels to operate. Lowering precision may be sufficient for earlier layers, but deeper layers that require a much more detailed representation would suffer from a loss of accuracy.

- **Pruning** is also another technique used to optimise energy efficiency mentioned in the same paper (Gorvadiya et al., 2025). This technique is a process where particular components in a neural network that may be problematic or simply are not as contributive are then removed in order to reduce the computational complexity for faster processing. Depending on what is being removed determines the type of pruning that is present, such as weight pruning, which is the removal of weights, and filter pruning, which is the removal of filters or kernels within the convolutional layers. To a significant extent, the computational intensity is reduced, and energy efficiency is increased. In most cases, after a round of pruning, another round may be necessary in order to maintain the accuracy. This must be done in an iterative manner after frequent retraining of the model to recover accuracy loss. Iterative pruning, retraining, and pruning again are crucial, as pruning a large amount in one go can significantly decrease accuracy. This results in computational overhead, producing more computational work and inevitably eliminating energy efficiency gains.

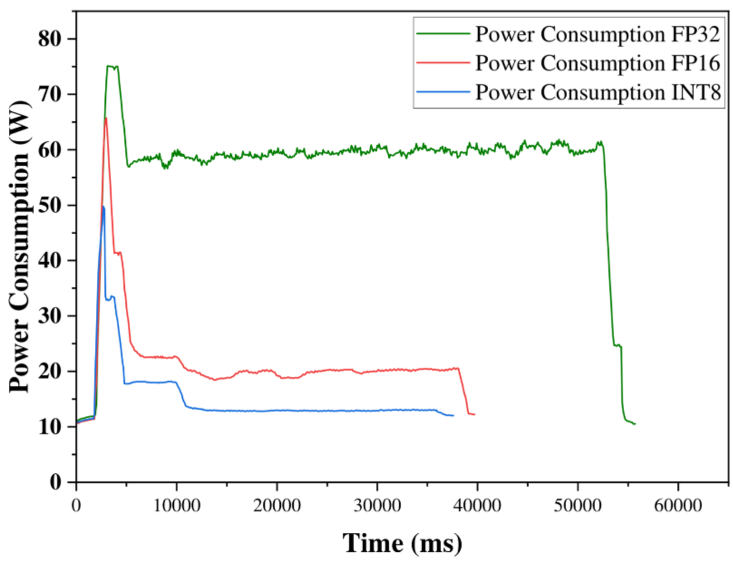

Figure 4: Impact of switching between INT8, FP16, and FP32 on Energy consumption for a single object detection (Gorvadiya et al., 2025)

2.2.3 *Systolic Arrays*

Inspired by the rhythmic nature of the human heart, which determined its name, the systolic array is a specialised architecture in the form of a grid of processing elements (PEs). Referring to figure 3, the systolic array is essentially the MAC array, which contains the PEs that perform computations. This core component within the neural processing unit allows it to perform millions of calculations in parallel as the PEs operate in a pipelined fashion. This reduces the energy consumption by a significant amount since the data flows locally from one PE to the next, involving no repetitive data movement to and from memory. Due to their structure, they are useful when it comes to computing matrix multiplications and convolution operations in CNNs, as shown in this equation (1). To maintain the structured flow of systolic array computations, internal memory components are utilised to help maintain data locality and efficient data transfer. Data from external memory, such as DRAM, is sent to a buffer memory directly connected to the systolic array, which will hold the necessary information to perform the correct calculations. In order for the PEs to perform those calculations, the buffer memory feeds data to internal temporary data registers of the first PE, where it will perform the computation with that data and pass that data onto the next PE with its internal register so that it does not have to fetch from primary memory. Referring to the equation (1) in this case, two matrices are being multiplied, and each PE will receive one element from each matrix and store them in the input registers to perform the dot-product operation. The data is temporarily held within that processing element before sending it to the next PE or storing it in the buffer at the end.

$$C[i][j] = \sum_{k} (A[i][k] \times B[k][j]) \tag{1}$$

Although the use of systolic arrays improve the energy efficiency and throughput of the NPU, they may still have challenges. In today's machine learning applications, there is a high demand for computational power while maintaining low energy consumption, especially based on the edge. Due to this demand, the complexity of the neural network model is increased, and so is the amount of data that needs to be processed. This is when further optimisation may be necessary.

- **Dataflow-Mirroring NPU**: It is evident that NPUs can suffer from low hardware utilisation, resulting in bottlenecks due to the systolic array efficiency being highly dependent on the characteristics of a given neural network. Spatial multitasking is proposed as a possible solution to improve this "utilisation". However, it falls short as it faces challenges due to the coarse-grained distribution of systolic arrays. This is essentially a way of grouping the PEs of the systolic array into large clusters where each block handles a significant computation. The solution proposed by (Choi et al., 2023) introduces fine-grained spatial multitasking, similar to the previous approach; this one has many small PEs to perform minimal computations (figure 5). The process NPU performs here is dataflow mirroring, where the DNN data flow is reversed either vertically or horizontally in order to optimise the allocation of PEs, ultimately improving the parallelism performance compared to a standard NPU. This approach improves performance by 35.1% over traditional NPU architectures.

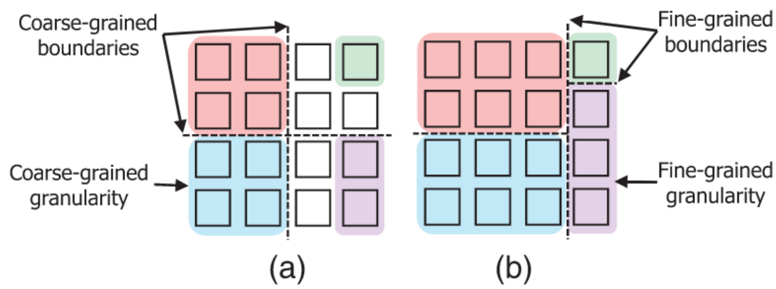

Figure 5: Difference between fine and coarse grain systolic array allocations (Choi et al., 2023)

- **Low-Precision Convolution Optimisation**:
  To make convolution operations work on a systolic array, the hardware architecture needs to be customised. A particular optimisation technique proposed by (S. Kim et al., 2021) is a channel-first order organisation, where it processes the spatial dimensions of the convolution kernel one

step at a time. By taking this approach of processing the kernel slices sequentially, it can fit into the systolic flow without extra memory usage. However, for this to work for multiple kinds of convolutions, it requires the accurate address of data to be stored in the array. To counter this challenge, the researchers designed a flexible address indexing unit that calculates the memory addresses for different convolutional operations. This ensures that the correct data is fed into the systolic array at the right time, regardless of kernel size.

- **Sparsity-aware Zero-skipping mechanism**:
  Systolic arrays offer a significant amount of parallelism and data flow. However, standard systolic arrays struggle with irregular parallelism when the data input may not be consistent or would have to deal with sparse data where the input is a zero. This leads to a breakage in the rhythmic flow that a systolic array would have and wastes computational resources.

  To address this issue, (S. Kim et al., 2021) proposed a design that integrates a zero-skipping mechanism into a system that is sparsity aware. By skipping the zero values, power consumption is minimal, and data transfer overhead is reduced. The design also incorporates a low-precision computation, so not only does the proposed architecture improve energy usage significantly, but the computational throughput still remains strong. From the mechanisms built into this accelerator, the results are observed to be an inference speed of 317 GOPS and an energy efficiency of 51.96 GOPS/W.

  A similar approach was taken by (Park et al., 2021) with a systolic array consisting of 6000 mac units enabling high parallelism. And with a reconfigurable adder-tree datapath as shown in figure 6 and figure 7, it reduces power consumption by 26% as it minimises energy associated with frequent flip-flop toggling. It also reduces computational load by skipping zeros during convolution operations by taking advantage of zero-value feature maps, resulting in an energy efficiency of 13.6 TOPS/W while maintaining 83.8% mac utilisation when active.

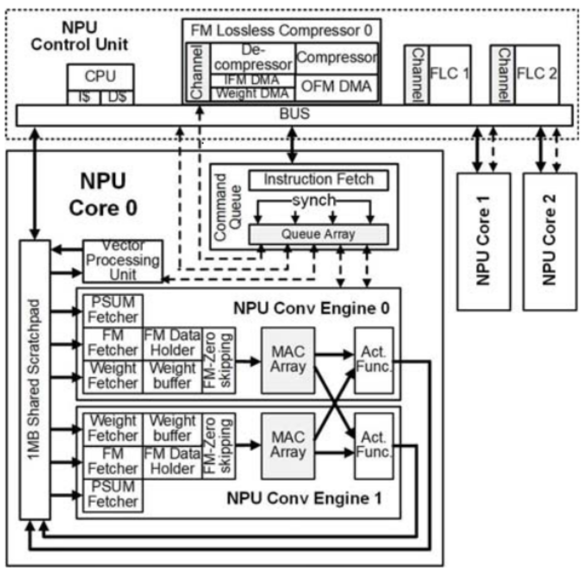

Figure 6: Neural Processing Unit Architecture

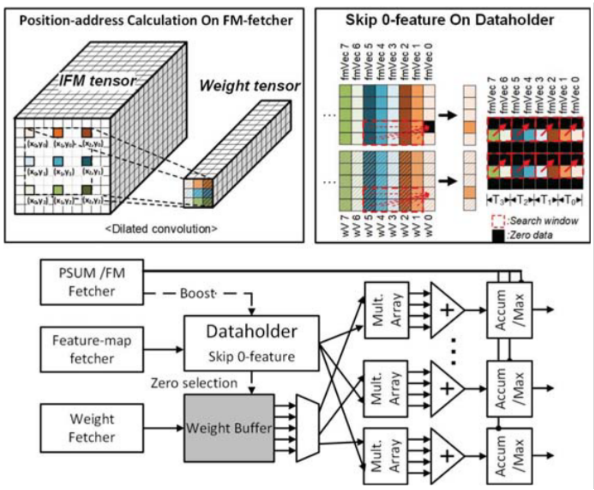

Figure 7: Fetching feature map and zero-skipping features on NPU

2.2.4 Energy Efficiency for edge

Day by day, artificial intelligence is shifting towards edge devices, where achieving energy efficiency is significantly difficult to accomplish. Unlike standard cloud-based systems, edge-based embedded systems are required to have real-time constraints for processing data alongside low energy consumption. Meeting these demands whilst maintaining low energy consumption is a substantial challenge, and even the most optimised NPUs still face issues. However, several promising approaches have been researched.

- **NeuroPipe**
  Conventional host-device execution models involves the CPU offloading neural-network computations to the NPU, which is fairly straightforward to manage but it leaves processing units idle at certain periods which can lead to wasted resources. While The proposed design by (B. Kim et al., 2020) involves the employment of a heterogenous processing approach to minimise idle cycles, the heterogenous architecture is not directly a relevant approach to this research focus. The power management techniques introduced within the system however, is presented to be a valuable consideration in order to improve the energy efficiency of a reconfigurable NPU in edge-based applications. Neuropipe's use of dynamic voltage and frequency scaling (DVfS) is a useful power management technique which comprises of a dynamic adjustment of the system voltage and frequency based on the computational load. It can monitor the inference workload and system performance by measuring throughput, latency, and energy consumption. Utilising the technique of DVFS in the context of a reconfigurable NPU would allow for an efficient inference across several neural network models by lowering the clock speed and voltage when the workload is light and ramp it up when it is high.

  Another technique this paper applies is offline and dynamic profiling to determine the suitable processing unit for each layer based on the task. In this research scope with a reconfigurable NPU only, implementing a similar technique can still be valuable, where the NPU could potentially make adjustments due to profiling the computational complexity of individual layers and dynamically adapting the dataflow configuration so that the NPU can leverage the concept of workload optimisation and power management without relying on multiple processors in a heterogeneous system.

- **PL-NPU**
  The research conducted from (Wang et al., 2022) proposes using a method called Posit-Based Logarithm-Domain Processing Elements (PLPE). This technique involves dynamically adapting to the data range and precision requirements during the training stage to reduce the complexity of MAC operations by using a low-bit-width posit format. The logarithmic domain computation converts the multiplication into additions, which reduces the power consumption of MAC operations. A hybrid floating-fixed-point accumulator (HFPA) is used to maintain high precision and reduce power usage by 35.6%. The research also utilises Inter-Intra-Channel-Reuse Dataflow (ICRD), a reconfiguration of the operand mapping pattern. During the neural network training phase, computations such as the feed-forward (FF), back-propagation (BP), and weight gradient cause a high frequency of data movement between the PEs and memory, leading to high energy consumption. The ICRD technique attempts to solve this issue by reconfiguring the operand mapping pattern so that data can be reused effectively with intra-channel and inter-channel operations. Hence, resultant values calculated from sections of the data can be reused within the same channel several times instead of fetching data continuously from SRAM, reducing memory access.

  Furthermore, a key component of the PL-NPU is the Pointed-Stake-Shaped Code Unit (PSCU). This is where operands are compressed into a an exponential-golomb format which is a code that represents small integers with shorter binary codes. This particular approach can be helpful, especially when dealing with sparse data and also reducing the actual amount of data that needs to be transferred between memory due to the encoding which takes up less space. As shown in figure 8, it demonstrates different components of the PL-NPU design results in more appealing values. The PL-NPU archives a peak energy efficiency of 3.87TFLOPS/W at 60Mhz with a power consumption of 343mW. PL-NPU also demonstrates 3.75 times higher energy efficiency and 1.68 times faster training of the ResNet18 model compared to state-of-the-art processors.

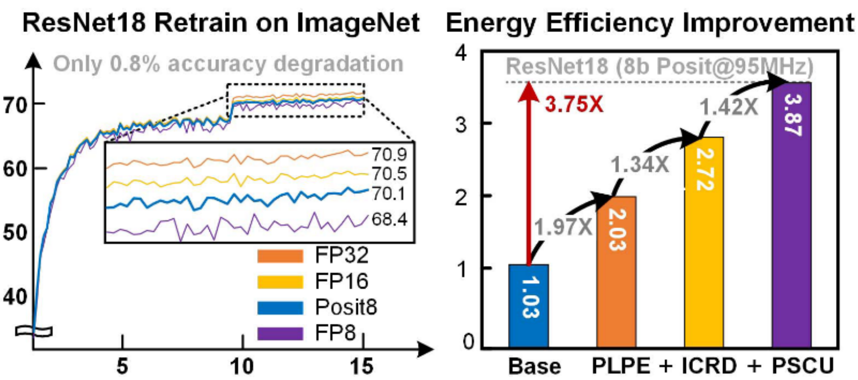

Figure 8: Energy Efficiency Improvement from each added method proposed in (Wang et al., 2022)

2.3 Field Programmable Gate Array (FPGA) Implementation

A Field Programmable Gate Array (FGPA) is a hardware platform that is highly customisable and exceptionally suitable for implementing an NPU. The reconfigurability allows for heavy control over the resources, which can be helpful when energy efficiency is a key requirement so that the architecture can be modified to produce a much more energy-efficient output from experimentation with low-precision logic, sparsity support and dataflow optimisations (S. Kim et al., 2021). They can also be reconfigured at run-time to balance performance and flexibility (X. Liu et al., 2024). There are several approaches to producing a reconfigurable NPU on the FPGA. The approach presented by (Guha & Chakrabarti, 2024) demonstrates the utilisation of DSP slices and LUTs much more adequately, achieving 388.18 GOPs/W. Their approach involves storing pre-computed values in the embedded memory blocks (EMBs), essentially removing the frequent memory fetching. Instead of calculating during the inference stage, this approach involves looking up the pre-computed values stored in the block RAM, reducing the power consumption, especially when it comes to matrix multiplication in CNN where there are repetitive mathematical operations.

Another informative approach to implementing an FPGA-based NPU was demonstrated by (Que et al., 2023). This approach utilises a multi-core architecture within the NPU, allowing it to switch between a single-core and multi-core operation depending on the tasks at hand. By using a thread-aware vector manager and buffer, it is able to switch tasks among threads when the neural networks come across data hazards, thus minimising potential idle periods and maintaining steady throughput. Compared to a single-threaded design, the Remarn architecture presents a significantly higher hardware utilisation. As mentioned in the research, when processing a long short-term memory (LSTM) RNN architecture, Remarn reaches up to 3.61 times the performance of the single-threaded application.

2.4 Research Gaps/Problem

- Finding balance between energy efficiency and accuracy when applying low-precision data representation
- Existing architectures, at times, react crudely from unstructured sparsity, leading to inefficient hardware utilisation.

- There is limited research when it comes to switching configurations within a single NPU based on the workload, especially when real-time requirements are involved, compared to heterogeneous multiprocessor systems.
- Developing an adaptive memory management technique which can dynamically allocate memory resources based on the task requirements.

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